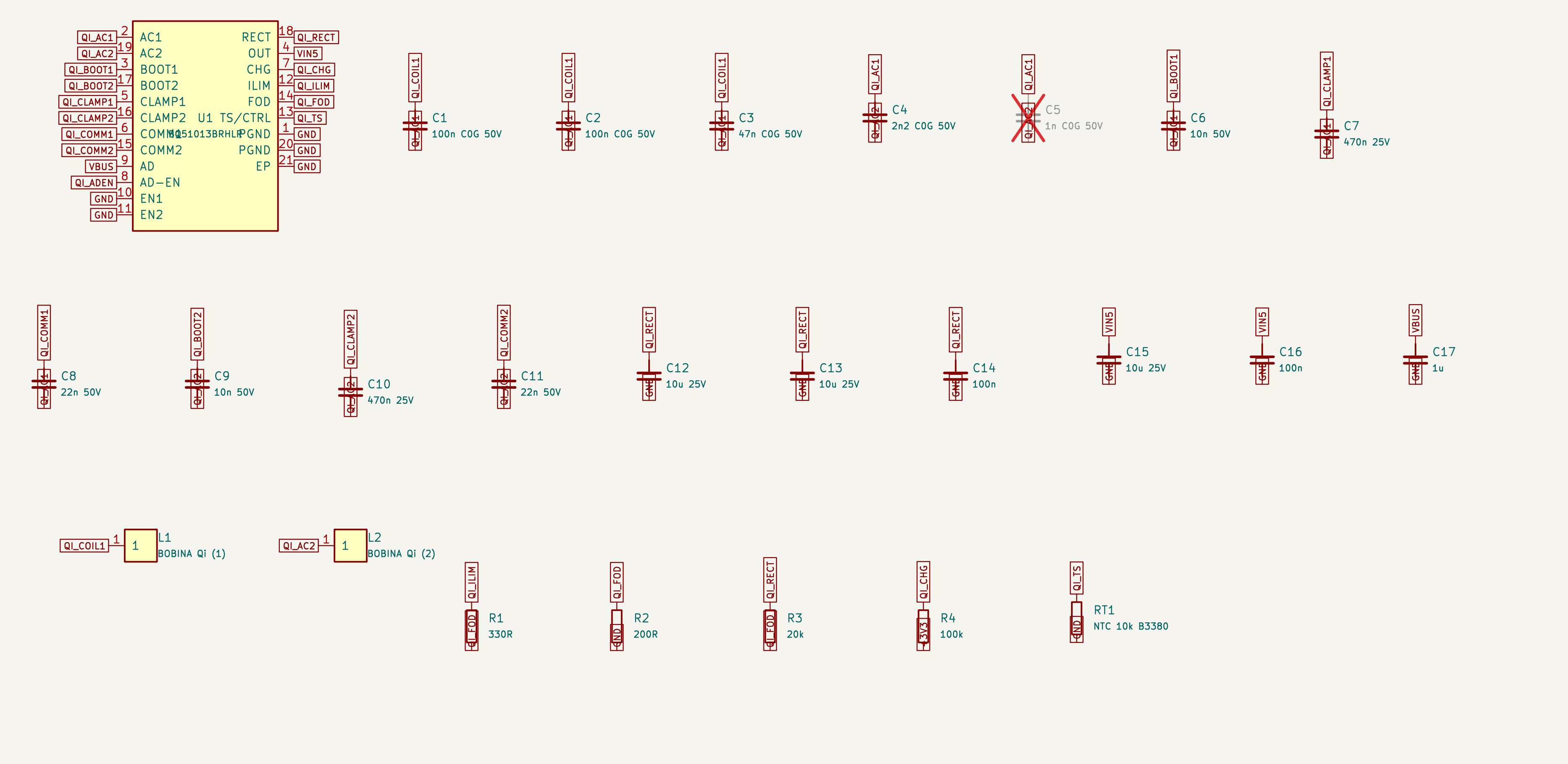
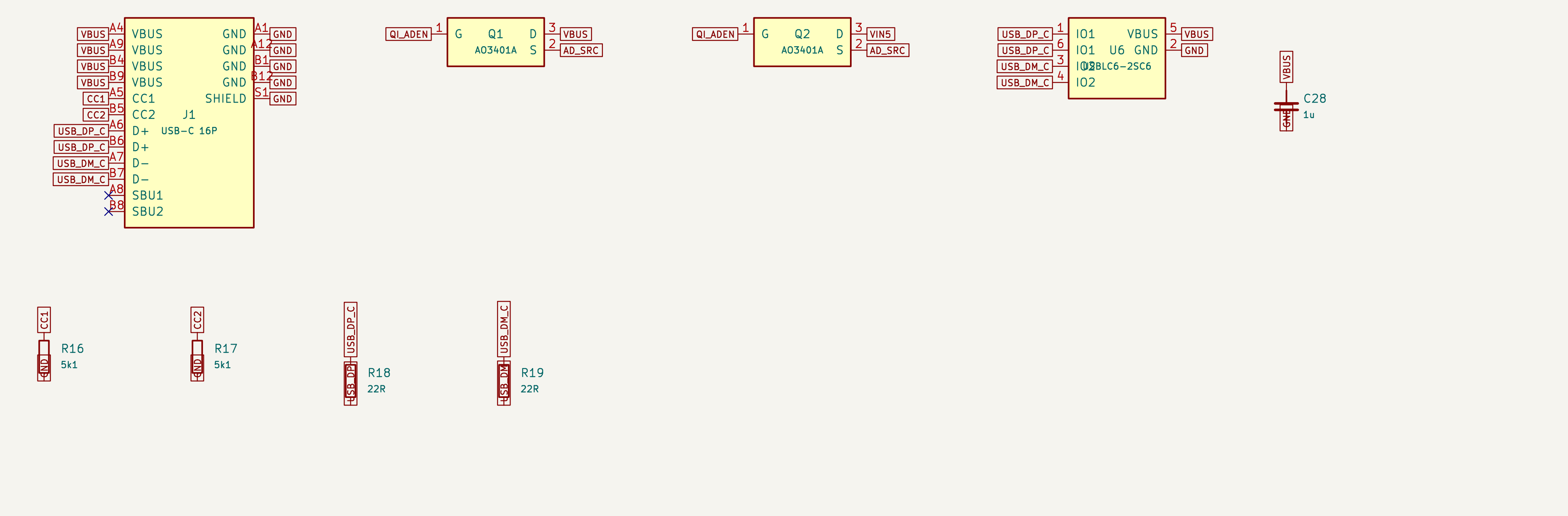
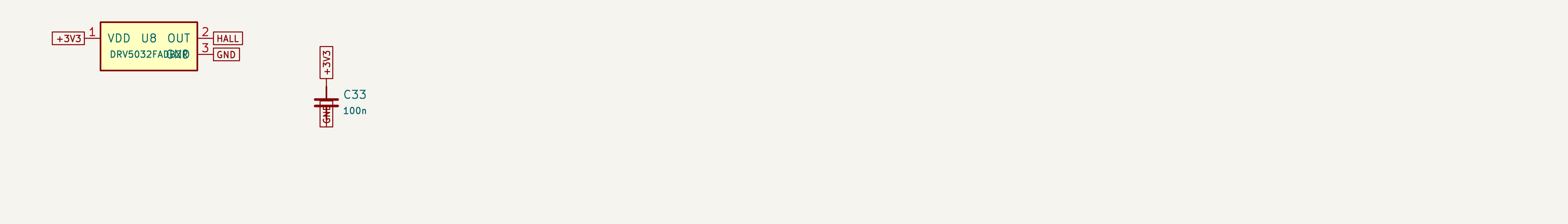
RECEPTOR Qi (BQ51013B) · bobina 30 × 23 bajo la placa · USB por AD/AD-EN


U1 BQ51013BRHLR
AC1
AC2
BOOT1
BOOT2
CLAMP1
CLAMP2
COMM1
COMM2
AD
AD-EN
EN1
EN2
RECT
OUT
CHG
ILIM
FOD
TS/CTRL
PGND
PGND
EP
C1 100n C0G 50V
C2 100n C0G 50V
C3 47n C0G 50V
C4 2n2 C0G 50V
C5 1n C0G 50V
C6 10n 50V
C7 470n 25V
C8 22n 50V
C9 10n 50V
C10 470n 25V
C11 22n 50V
C12 10u 25V
C13 10u 25V
C14 100n
C15 10u 25V
C16 100n
C17 1u
L1 BOBINA Qi (1)
L2 BOBINA Qi (2)
R1 330R
R2 200R
R3 20k
R4 100k
RT1 NTC 10k B3380

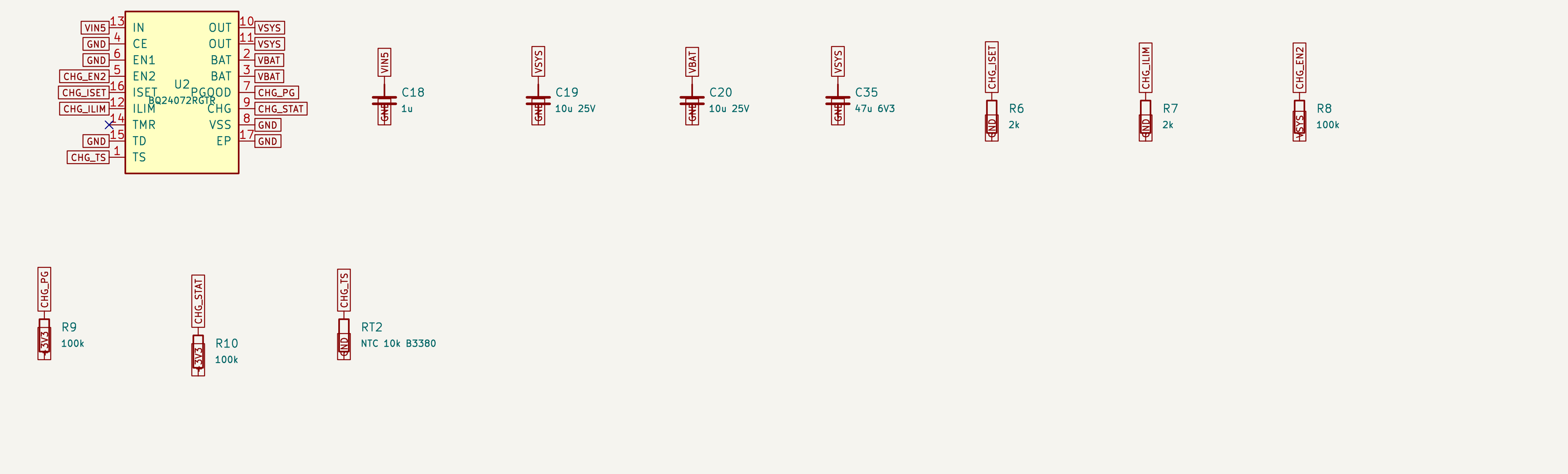
USB-C (programación y carga) · ESD


J1 USB-C 16P
VBUS
CC1
CC2
D+
D-
SBU1
SBU2
GND
SHIELD
Q1 AO3401A
Q2 AO3401A
U6 USBLC6-2SC6
C28 1u
R16 5k1
R17 5k1
R18 22R
R19 22R

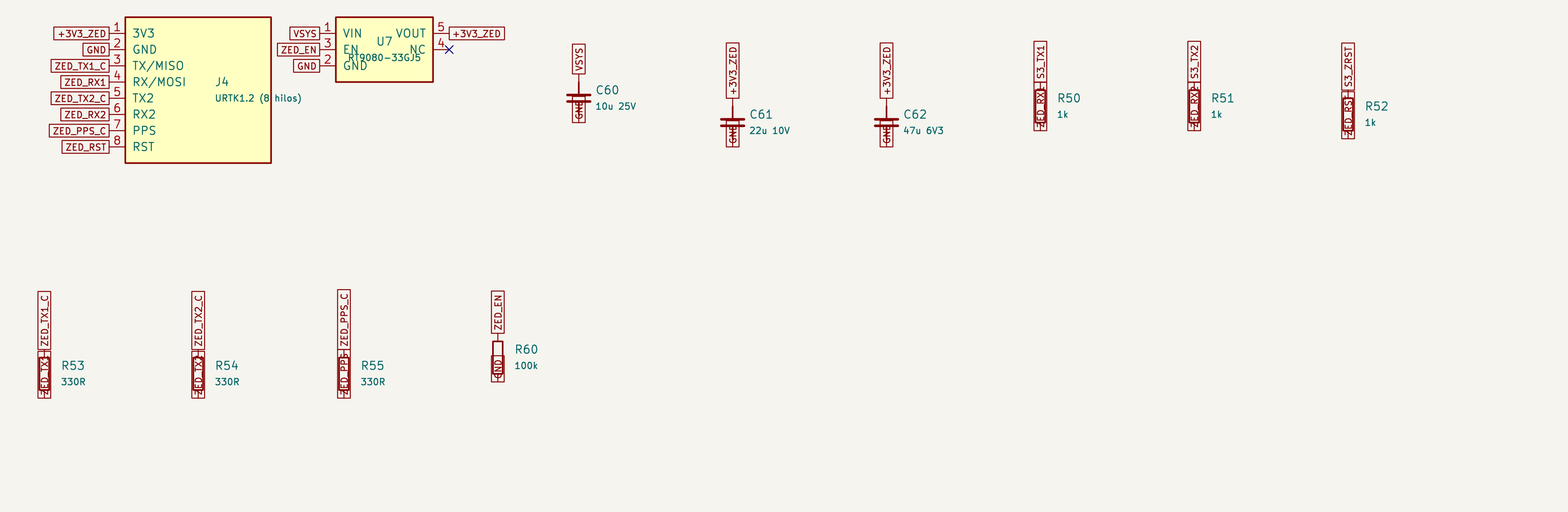
SENSOR DE SOPORTE: Hall DRV5032FA (imanes del soporte)


U8 DRV5032FADBZR
VDD
OUT
GND
HALL
C33 100n

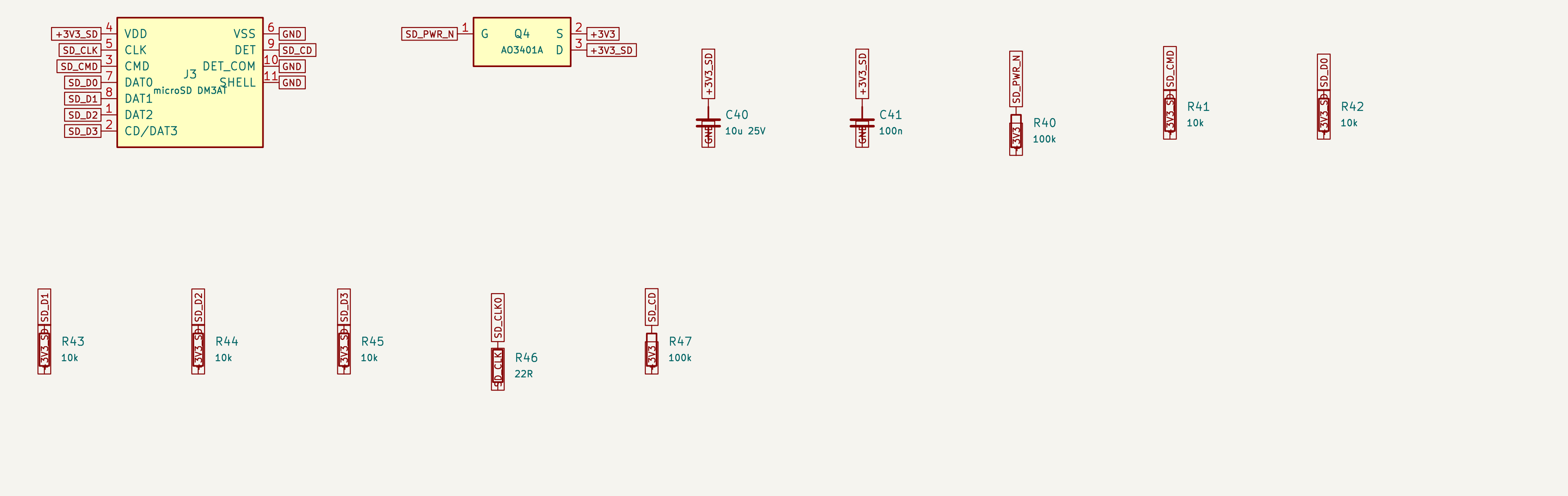
CARGADOR CON CAMINO DE POTENCIA (BQ24072) · batería 1S 2000 mAh (654060)


U2 BQ24072RGTR
IN
CE
EN1
EN2
ISET
ILIM
TMR
TD
TS
OUT
BAT
PGOOD
CHG
VSS
EP
C18 1u
C19 10u 25V
C20 10u 25V
C35 47u 6V3
R6 2k
R7 2k
R8 100k
R9 100k
R10 100k
RT2 NTC 10k B3380

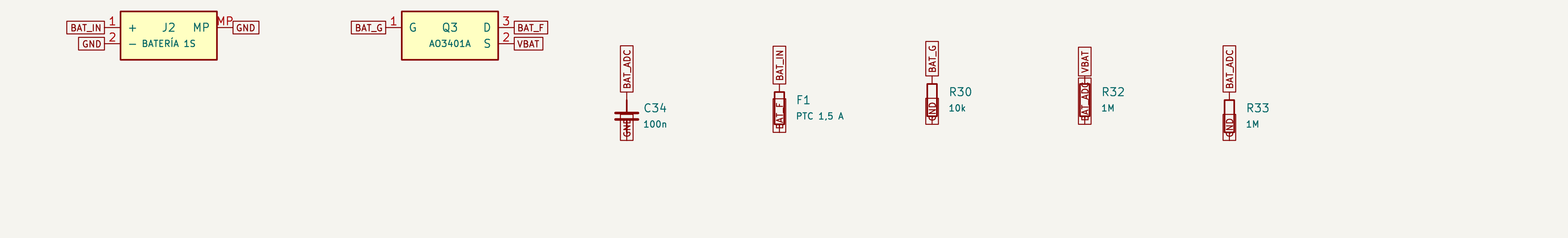
URTK1.2 / ZED-F9P: 3,3 V propio con EN (RT9080) · UART1 · UART2 · PPS · RESET


J4 URTK1.2 (8 hilos)
3V3
GND
TX/MISO
RX/MOSI
TX2
RX2
PPS
RST
U7 RT9080-33GJ5
VIN
EN
GND
VOUT
NC
C60 10u 25V
C61 22u 10V
C62 47u 6V3
R50 1k
R51 1k
R52 1k
R53 330R
R54 330R
R55 330R
R60 100k

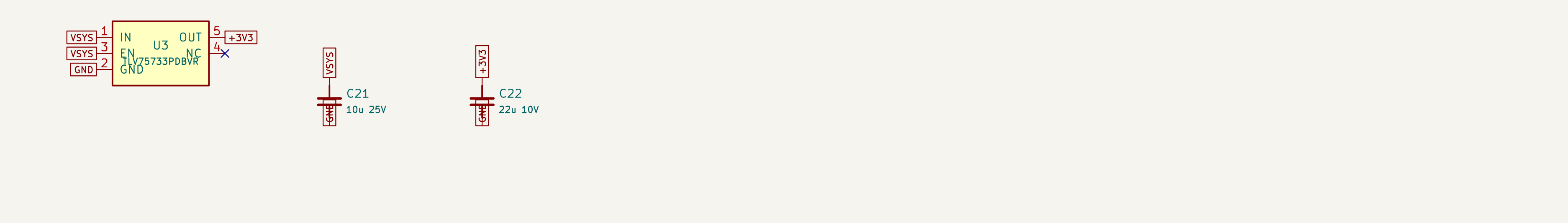
microSD (SDMMC 4 bits) · alimentación conmutada


J3 microSD DM3AT
VDD
CLK
CMD
DAT0
DAT1
DAT2
CD/DAT3
VSS
DET
DET_COM
SHELL
Q4 AO3401A
C40 10u 25V
C41 100n
R40 100k
R41 10k
R42 10k
R43 10k
R44 10k
R45 10k
R46 22R
R47 100k

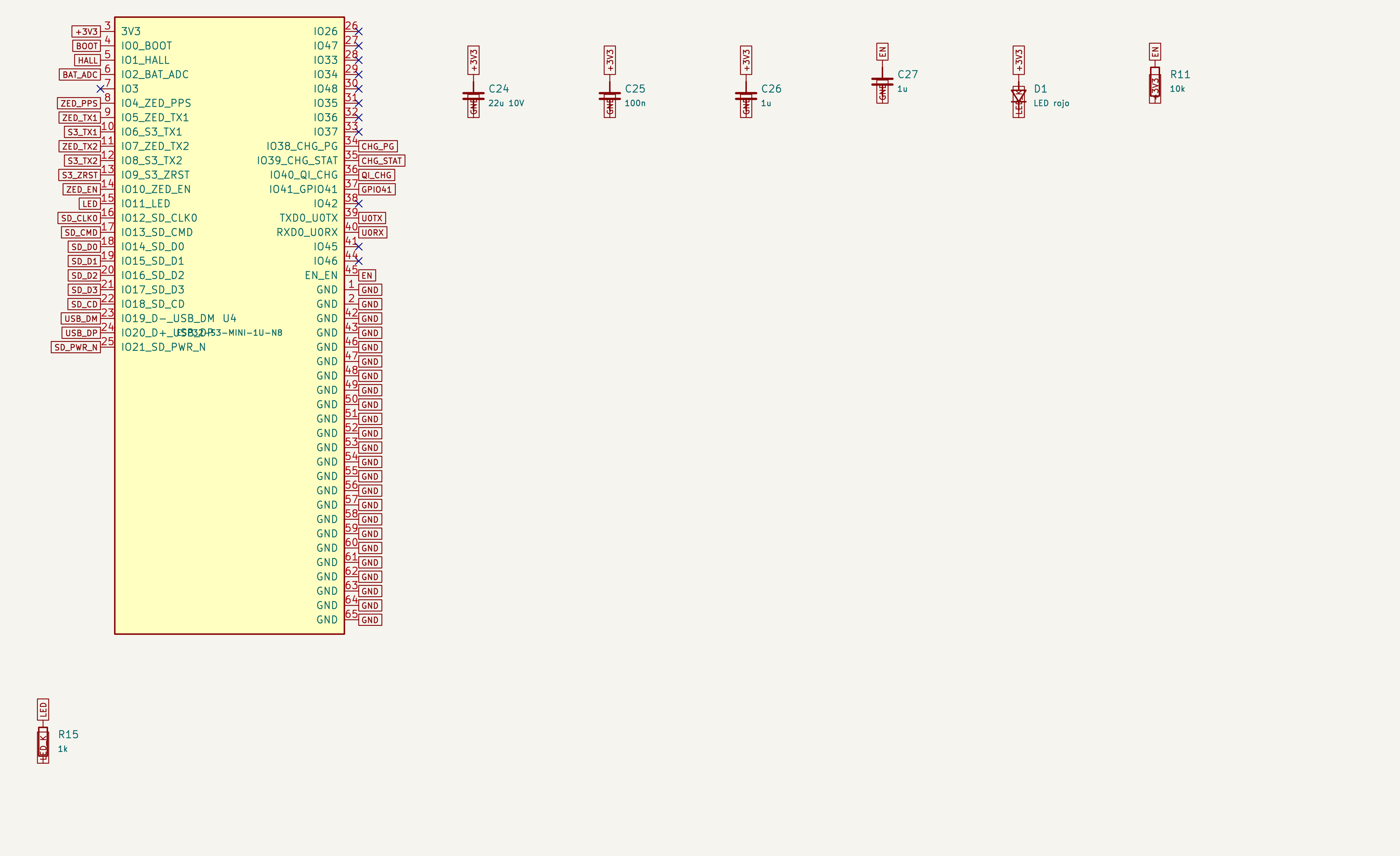
BATERÍA: PTC + protección de polaridad · medida por ADC


J2 BATERÍA 1S
Q3 AO3401A
C34 100n
F1 PTC 1,5 A
R30 10k
R32 1M
R33 1M

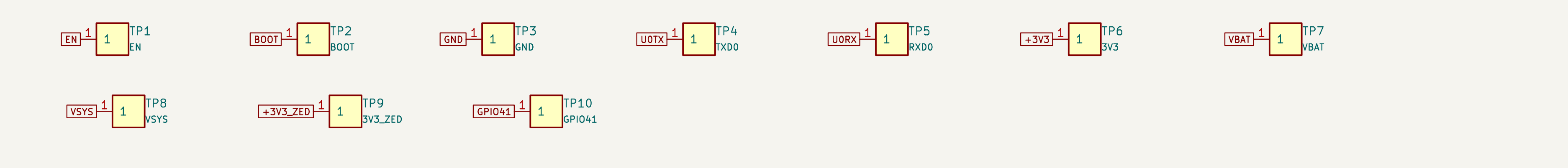
3,3 V PRINCIPAL (TLV75733P, 1 A)


U3 TLV75733PDBVR
IN
EN
GND
OUT
NC
C21 10u 25V
C22 22u 10V

ESP32-S3-MINI-1U-N8 · arranque · LED


U4 ESP32-S3-MINI-1U-N8
3V3
IO0_BOOT
IO1_HALL
IO2_BAT_ADC
IO3
IO4_ZED_PPS
IO5_ZED_TX1
IO6_S3_TX1
IO7_ZED_TX2
IO8_S3_TX2
IO9_S3_ZRST
IO10_ZED_EN
IO11_LED
IO12_SD_CLK0
IO13_SD_CMD
IO14_SD_D0
IO15_SD_D1
IO16_SD_D2
IO17_SD_D3
IO18_SD_CD
IO19_D-_USB_DM
IO20_D+_USB_DP
IO21_SD_PWR_N
IO26
IO47
IO33
IO34
IO48
IO35
IO36
IO37
IO38_CHG_PG
IO39_CHG_STAT
IO40_QI_CHG
IO41_GPIO41
IO42
TXD0_U0TX
RXD0_U0RX
IO45
IO46
EN_EN
GND
C24 22u 10V
C25 100n
C26 1u
C27 1u
D1 LED rojo
R11 10k
R15 1k

PUNTOS DE PRUEBA


TP1 EN
TP2 BOOT
TP3 GND
TP4 TXD0
TP5 RXD0
TP6 3V3
TP7 VBAT
TP8 VSYS
TP9 3V3_ZED
TP10 GPIO41

Generado por fuente/avo_back_v4_pcb.py (mismo origen de datos que la placa). Las conexiones son etiquetas globales: dos pines con la misma etiqueta están unidos.

AVO Performance Horse
Sheet: /
File: AVO_BACK_V4_PCB.kicad_sch
Title: AVO BACK V4 · placa de la espalda · REV A
Size: User
Date: 2026-10-29
Rev: A
KiCad E.D.A. 9.0.9
Id: 1/1